

LEENA KANNAN

Chennai, India | leenakannan.23@gmail.com | 8248529608 | [LinkedIn](#) | [LeetCode](#)

SUMMARY

AI Engineer and final-year ECE student with hands-on experience building production-grade AI systems across LLMs, voice agents, and real-time ML pipelines. Worked on enterprise RAG architectures, agent orchestration, and observability-heavy ML systems, with recent focus on conversational voice agents, TTS fine-tuning, and telephony infrastructure. Strong foundations in Python, C++, PyTorch, distributed systems, and systems-level thinking.

PROFESSIONAL EXPERIENCE

Tech Intern, Emvo.AI Nov 2025 - Present

- Built and integrated real-time conversational voice agents, connecting LLM reasoning, ASR, and fine-tuned TTS pipelines for low-latency production deployment.
- Implemented and debugged FreeSWITCH-based telephony infrastructure for AI-driven inbound/outbound calls, handling call routing, audio streaming, and system-level failures.
- Contributed to improving end-to-end voice latency and reliability across model inference and telephony layers.

Artificial Intelligence Intern, Zoho Corporation Pvt. Ltd. | [GitHub](#) May 2025 - Jun 2025

- Architected and deployed production-grade AI assistant with RAG architecture, integrating multiple LLMs (GPT-4o, Gemini, Groq) and orchestration frameworks (LangChain, LlamaIndex, AutoGen, CrewAI).
- Designed flexible vector database layer supporting FAISS, ChromaDB, and Qdrant for high-speed semantic search and domain-specific knowledge retrieval.
- Established comprehensive observability stack using OpenTelemetry, Prometheus, and Site24x7 for real-time tracing, usage analytics, and cost optimization.

PROJECTS

Multi-Future Visual Perception via Probabilistic State Superposition for Robotics Dec 2025 - Present

- Designed an uncertainty-aware robotic perception framework that maintains a belief distribution over multiple plausible scene states instead of a single deterministic prediction.
- Implemented probabilistic state evolution and hypothesis updating to anticipate future object dynamics under partial observability and noisy visual inputs.
- Evaluated the framework in robotic perception tasks such as object motion prediction and interaction planning, demonstrating more robust decision-making than conventional deterministic pipelines.

Real-Time Voice Agent Evaluation & Latency Benchmarking Tool | [GitHub](#) Jun 2025 - Sep 2025

- Built a latency benchmarking and evaluation system for conversational voice agents, measuring end-to-end response time across ASR, LLM reasoning, and TTS pipelines.
- Instrumented voice pipelines to isolate performance bottlenecks in audio streaming, model inference, and telephony integration.
- Enabled data-driven optimization of production voice agents by providing repeatable metrics and comparative evaluations across different model and infrastructure configurations.

Materials Property Predictor | [GitHub](#) Jan 2025 - Apr 2025

- Built an end-to-end production ML system using PyTorch to predict material properties from chemical composition, supporting rapid screening and discovery workflows.
- Engineered a feature extraction pipeline using Matminer (132+ compositional features) and Monte Carlo dropout for uncertainty quantification, achieving $R^2 > 0.92$.
- Deployed a containerized microservices architecture with FastAPI, Streamlit, and Redis caching, reducing prediction latency by ~60%.

SKILLS

Languages: Python, C++

Frameworks: FastAPI, Flask, PyTorch, Scikit-Learn, Streamlit, Numpy, Pandas

AI/LLM: LangChain, LlamaIndex, AutoGen, CrewAI, spaCy, NLTK, Google ADK, PipeCat, STT, TTS

Infra & DevOps: Docker, Docker Compose, Nginx, Prometheus, OpenTelemetry, CI/CD, MLOps

Tools: Google Colab, Git, GitHub, MLFlow, Weights & Biases

Strengths: Distributed Systems, Observability, ML Pipelines, System Monitoring, Cross-functional Collaboration.

EDUCATION

Bachelor of Technology, Electronics and Communications Engineering

Sep 2022 - May 2026

VIT University, Vellore : CGPA 8.35

ADDITIONAL INFORMATION

- Certifications: AWS Certified Cloud Practitioner (Aug 2024 - Aug 2027)
Credential ID: f53e8af1662a4104a38fa1ecbbad5772
- Awards/Activities: Co-Secretary of Society for Industrial and Applied Mathematics (Aug 2025 - Present), Winner - Vellore District Women's Football Tournament (2023)